Survivor Loss and Fuzzy Cutvertices: Mathematical Modelling of Key-Person Risk in Informal Communication Networks

Supplementary material to the article of the same title. Section, definition and proposition numbers refer to the main article.

Supplementary Material. Proofs and Implementation Details

S1 Proof of Proposition 4.1

Proof. (1) $\sigma(v) \in [0,1]$ because $0 \le a_v \le a_{max}$ and $\ln(1 + x)$ is increasing. Since $0 \le 1 - e^{-\lambda w} < 1$, $\mu(u,v)$ never exceeds $\sigma(u) \wedge \sigma(v)$, and μ is symmetric because $w_{uv} = w_{vu}$. (2) If u and v communicated, then $w_{uv} > 0$ and $a_u, a_v > 0$, so $\sigma(u), \sigma(v) > 0$ and $\mu(u,v) > 0$; otherwise $w_{uv} = 0$ and $\mu(u,v) = 0$. (3) With σ fixed, $1 - e^{-\lambda w}$ is strictly increasing in w. ∎

Part (3) is stated for fixed σ because σ itself depends on the link weights through a_v and a_{max}: raising one w_{uv} can change the memberships of u, v and, through a_{max}, of other employees.

S2 General Form of Corollary 4.1

For a departing set $S \subset V$, the surviving-pair capacity before departure follows by inclusion–exclusion:

$$Q_{\text{surv}}(G;S) = Q(G) - \sum_{w \in S} \sum_{x \neq w} \text{CONN}_G(w,x) + \sum_{\substack{w \prec w' \\ w,w' \in S}} \text{CONN}_G(w,w').$$

The first sum removes every pair with at least one departing member, counting pairs with both members in S twice; the second sum adds those back once. $Q(G - S)$ is obtained by applying Proposition 4.2 to $G - S$.

S3 Algorithm 1 and Implementation

Algorithm 1 (fuzzy bridges and hidden key employees, from Proposition 4.3):

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Input : fuzzy graph G = (σ, μ); set M of all distinct positive arc memberships
(exact values)
Output: fuzzy bridges B, fuzzy cutvertices C, hidden key employees H

1  B ← ∅ ; C ← ∅
2  for each t in M (descending):          # every distinct positive level is
visited
3      H_t ← crisp graph on V with arcs {(u,v) : μ(u,v) ≥ t}
4      run Tarjan's bridge / articulation-point search on H_t
5      B ← B ∪ {bridges (u,v) of H_t with μ(u,v) = t}
6      C ← C ∪ {articulation points of H_t}
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7  H ← C \ {articulation points of G*}    # C is the union over all t in M
8  return B, C, H
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Each iteration uses Tarjan's (1972) bridge and articulation-point search. The exact membership values are used: rounding μ creates ties, which add equally strong alternative paths and change which vertices are fuzzy cutvertices. With exact values almost every arc has its own level, so |M| can approach the number of arcs; the following lemma and Proposition 3.2 reduce the work.

Lemma S.1. Every fuzzy bridge of G belongs to every maximum spanning forest of G*.

Proof. Let e = (u,v) be a fuzzy bridge and suppose some maximum spanning forest F does not contain e. By Lemma 4.1, the u–v path in F has strength $\mathrm{CONN}_G(u,v) \geq \mu(e)$, and it avoids e. Hence $\mathrm{CONN}_{G-e}(u,v) \geq \mu(e)$, contradicting Theorem 3.1. ∎

Implementation. (i) Fuzzy bridges are searched only among the at most n − 1 arcs of one maximum spanning forest F: an arc e = (u,v) of F is a fuzzy bridge exactly when u and v are disconnected in $H_{\mu(e)} - e$ (Proposition 4.3(1)). (ii) By Proposition 3.2, the fuzzy cutvertices are exactly the employees with KS(w) > 0, which are computed in any case; Algorithm 1's cutvertex output is run as an independent check, and the two sets must coincide on every dataset. Both shortcuts, together with Algorithm 1, were verified on 150 random fuzzy graphs (half with tied memberships), where every result matched a brute-force evaluation of the definitions (verify.py in the code repository). (iii) The mathematical criterion is KS(w) > 0; because floating-point arithmetic can leave residual values at machine precision, the implementation treats $\mathrm{KS}(w) > 10^{-12}$ as positive.